Haoyang Chen

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Research Interests

My research investigates how people can understand and steer AI decision-making through interactive explanations. I develop interfaces and learning methods that translate human feedback into changes in agent behavior.

Education

National University of Singapore Aug 2024 – Present
PhD Candidate, School of Computing

Southeast University Sep 2020 – Sep 2024
Bachelor of Computer Science

Publications

*Corresponding author

[4] Y. Wang, Q. Sun, **H. Chen***. 2026.
TreePIN^2: Tree-Prior Initialized Neural Networks.
Preprint. [Paper] [Code]

[3] **H. Chen**, J. Bai, T. Fang, and B. Lim. 2026.
Editable XAI: Toward Bidirectional Human-AI Alignment with Co-Editable Explanations of Interpretable Attributes.
ACM Conference on Human Factors in Computing Systems (*CHI*). [Paper] [Code]

[2] **H. Chen**, P. Sun, Q. Song, W. Wang, W. Wu, W. Zhang, G. Gao, and Y. Lyu. 2024.
i-Rebalance: Personalized Vehicle Repositioning for Supply Demand Balance.
AAAI Conference on Artificial Intelligence (*AAAI*). [Paper] [Code]

[1] **H. Chen**, Z. Li, and Y. Yao. 2022.
Multi-agent Reinforcement Learning for Fleet Management: A Survey.
International Conference on Artificial Intelligence, Automation, and High-Performance Computing (*AIAHPC*). [Paper]

Research Experience

PhD Candidate, *Ubicomp Lab, NUS* Aug 2024 – Present
Advisor: Prof. Brian Lim

Research Assistant, *Ubicomp Lab, NUS* Aug 2023 – Apr 2024
Advisor: Prof. Brian Lim

Research Assistant, *COOLA Lab, Southeast University* Aug 2021 – Aug 2023
Advisors: Prof. Yan Lyu and Prof. Wanyuan Wang

Honors & Awards

Research Achievement Award AY2025/26
School of Computing, National University of Singapore

Meritorious Winner May 2022
Interdisciplinary Contest in Modeling (ICM)

Tencent Scholarship Nov 2021
Top 10% of students

Teaching

National University of Singapore

CS4246/5446 · Reinforcement Learning and Sequential Decision Making AY2026/27, Sem 1

CP2107 · Independent Introduction to CS Research (Odyssey) May–Aug 2026

CS3263 · Foundations of Artificial Intelligence AY2025/26, Sem 2

CS3244 · Machine Learning AY2025/26, Sem 1

Professional Service

Reviewer

ACM Conference on Human Factors in Computing Systems (CHI) 2026, 2027

AAAI Conference on Artificial Intelligence (AAAI) 2027

IEEE Transactions on Intelligent Transportation Systems 2025